Structural Integrity of an Ice Shell Concentric Torus Habitat in Arcadia Planitia, Mars

Concept-level calculation

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Abstract

This note checks whether a self-supporting ice vault (one ring of a torus habitat) can stand for spans of 16 m, 24 m, 32 m, and 40 m, and how much heat from the habitat affects the ice. Compressive stress in the ice is small (46–115 kPa), but creep-driven buckling sets the shell thickness: 0.6 m for 16–24 m spans, 1.25 m for 32 m and 3.4 m for 40 m. The governing variable is not span but ice temperature: the shell must stay near −60 °C, with insulation on the inner face ($R \geq 4\,\mathrm{m^2\,K/W}$) and a reflective cover on the outer face. All numbers are concept-level estimates.

1 Summary of results

Table 1: Summary by span. FS = buckling safety factor after 30 years of creep. "Damaged" = reflective cover lost (§8.4).

L (m)	h (m)	t (m)	$\sigma_{\max}$ (kPa)	FS design	FS damaged	Arch mass (t)	Tie force (kN)
16	9.0	0.60	46	23.6	2.93	35	23
24	13.5	0.60	69	3.3	0.39	53	35
32	18.0	1.25	92	2.6	0.31	146	97
40	22.5	3.40	115	2.5	0.37	498	330

Arch mass and floor-tie force are per 2.5 m-wide segment.

2 Data and assumptions

Quantity	Value	Note
Mars gravity g	$3.71\,\mathrm{m/s^2}$	
Ice density ρ	$917\,\mathrm{kg/m^3}$	bubble-free
Ice elastic modulus E_0	9 GPa	short-term
Latent heat L_f	$334\,\mathrm{kJ/kg}$	
Specific heat c_p	$1{,}700\,\mathrm{J/(kg\,K)}$	near −60 °C
Conductivity $k(T)$	$9.828\,e^{-0.0057T}$ W/(m K)	T in K; $k(-60\,^\circ\mathrm{C}) = 2.92$ W/(m K)
Habitat air T_{in}	20 °C	
Mean outer-face temperature T_o	−60 °C	with reflective cover; seasonal swing ±15 °C
Insulation, module to ice R_{ins}	$4\,\mathrm{m^2\,K/W}$	e.g. aerogel blanket ≈ 6 cm
Rise/span ratio h/L	0.5625	same as the 16 m × 9 m arch
Design life t_d	30 years	
Minimum thickness (shielding)	0.6 m	$\approx 55\,\mathrm{g/cm^2}$

The 56.5 kPa habitat pressure is held by inflatable modules, not by the ice. Between module and shell is a gap of low-pressure Mars gas, so the ice carries only its own weight.

3 Arch geometry

The arch is an inverted *catenary*. Under uniform self-weight along the arch, this shape puts the whole section in pure compression, with no bending:

$$y(x) = h - a\left[\cosh\frac{x}{a} - 1\right], \qquad -\tfrac{L}{2} \le x \le \tfrac{L}{2}. \tag{1}$$

The parameter a follows from the feet sitting on the ground:

$$a\left[\cosh\frac{L}{2a} - 1\right] = h. \tag{2}$$

Arc length:

$$s = 2a\sinh\frac{L}{2a}. \tag{3}$$

4 Loads

The main load is self-weight per unit shell area:

$$w = \rho g t. \tag{4}$$

Other loads are very small:

- **Wind.** Dynamic pressure $q = \frac{1}{2}\rho_{atm}v^2 \approx \frac{1}{2}(0.02)(30)^2 \approx 9\,\text{Pa}$, under 1 % of the weight of a 0.6 m shell ($\approx 2.0\,\text{kPa}$). Neglected.
- **Habitat pressure.** Carried by the inflatable modules; it never reaches the ice.
- **Marsquakes.** Very weak compared with earthquakes. Neglected at concept stage.

5 Compressive stress

For a catenary, the horizontal force is constant, $H = w\,a$, and the axial force at height y is $N = w\,(a + h - y)$. It peaks at the feet ($y = 0$). Dividing by thickness t, the thickness cancels:

$$\boxed{\sigma_{\max} = \rho\, g\, (a + h)} \tag{5}$$

So the compressive stress depends only on arch size, not thickness. The crushing strength of cold ice is taken as 5 MPa (conservative).

Table 2: Geometry and compressive stress.

L (m)	h (m)	a (m)	s (m)	$\sigma_{\max}$ (kPa)	FS crushing
16	9.0	4.56	25.5	46.1	108
24	13.5	6.85	38.3	69.2	72
32	18.0	9.13	51.1	92.3	54
40	22.5	11.41	63.9	115.4	43

The ice will not crush: the safety factor exceeds 40 for every span.

6 Ice creep

Ice under sustained stress flows slowly. We use Glen's law ($n = 3$):

$$\dot{\varepsilon}_{ij} = A(T)\,\tau_e^{\,n-1}\,\tau_{ij}. \tag{6}$$

For uniaxial compression σ: $\tau_{11} = \tfrac{2}{3}\sigma$ and $\tau_e = \sigma/\sqrt{3}$, so

$$\dot{\varepsilon} = \tfrac{2}{9}\,A(T)\,\sigma^3. \tag{7}$$

The rate factor A depends strongly on temperature (Arrhenius; Cuffey & Paterson 2010):

$$A(T) = A^*\exp\!\left[-\frac{Q_c}{R}\left(\frac{1}{T}-\frac{1}{T^*}\right)\right], \quad A^* = 3.5\cdot10^{-25}/(\mathrm{s\,Pa^3}),\ T^* = 263\,\mathrm{K},\ Q_c = 60\,\mathrm{kJ/mol}. \tag{8}$$

Values: $A(-60\,^\circ\mathrm{C}) = 5.63\cdot10^{-28}$, $A(-40\,^\circ\mathrm{C}) = 1.03\cdot10^{-26}$, $A(-20\,^\circ\mathrm{C}) = 1.18\cdot10^{-25}$ $1/(\mathrm{s\,Pa^3})$. Every 20 °C of warming makes ice creep roughly 10–20 times faster.

Total creep strain over the design life, and the effective modulus used for buckling:

$$\varepsilon_c = \tfrac{2}{9}\,\bar{A}\,\sigma_{\max}^3\,t_d, \qquad E_{\mathrm{eff}} = \frac{\sigma_{\max}}{\sigma_{\max}/E_0 + \varepsilon_c}. \tag{9}$$

$\bar{A}$ is A averaged through the shell thickness (temperature varies linearly from inner to outer face) and over one year (sinusoidal seasonal swing). Because A rises exponentially with temperature, the warmest part dominates.

7 Buckling

A thin arch can snap out of shape and collapse even at low stress. For a two-hinged parabolic/catenary arch under uniform load (Timoshenko & Gere 1961), the critical load per metre of width is

$$q_{cr} = \gamma\,\kappa\,\frac{E_{\mathrm{eff}}\,I}{L^3}, \qquad I = \frac{t^3}{12}, \tag{10}$$

with $\gamma \approx 36$ for $h/L \approx 0.56$ and an imperfection factor $\kappa = 0.5$. The self-weight load per horizontal metre is $q = \rho g t\, s/L$. The safety factor is

$$\mathrm{FS} = \frac{q_{cr}}{q} = \frac{\gamma\,\kappa\,E_{\mathrm{eff}}\,t^2}{12\,\rho\,g\,s\,L^2} \quad\Longrightarrow\quad t_{\mathrm{req}} = L\sqrt{\frac{12\,\mathrm{FS}\,\rho\,g\,s}{\gamma\,\kappa\,E_{\mathrm{eff}}\,L}}. \tag{11}$$

Target FS ≥ 2.5. The calculation is iterated because E_{eff} depends on temperature, and the inner-face temperature depends on thickness (§8). Design thickness is rounded up to 5 cm and is never below 0.6 m (radiation shielding).

Table 3: Required thickness against buckling (design case: $T_o = -60\,^\circ\mathrm{C} \pm 15\,^\circ\mathrm{C}$, $R_{ins} = 4$).

L (m)	t_{req} (m)	t design (m)	ε_c 30 yr (%)	E_{eff} (GPa)	FS
16	0.18	0.60	0.004	0.97	23.65
24	0.51	0.60	0.014	0.46	3.31
32	1.22	1.25	0.046	0.20	2.58
40	3.39	3.40	0.229	0.05	2.50

Required thickness grows much faster than span: from 24 m to 40 m it rises more than fivefold. Two reasons: a larger arch loads the ice harder, so it creeps faster (E_{eff} drops), and buckling stiffness falls with L^3. Stiffening ribs at segment joints would reduce this, because I grows with the cube of section depth.

8 Thermal behaviour and inner-surface melting

8.1 Heat-flow model

Heat flows from the habitat air, through the insulation, through the ice, and is radiated to the sky from the outer face. Steady state, one dimension:

$$q = \frac{T_{in} - T_o}{R_{ins} + t/k}, \qquad T_i = T_o + q\,\frac{t}{k}, \tag{12}$$

where q is the heat flux (W/m^2) and T_i the inner-face ice temperature. Total heat loss for ring 1 uses the inner area $A_{in} = 2\pi R_1 s$ with $R_1 = 20\,\text{m} + L/2$ (Ø40 m courtyard). This is an upper bound, since the modules are smaller than the shell.

8.2 Melting criterion

The inner face reaches 0 °C if the flux into the ice exceeds

$$q_{\text{melt}} = \frac{k\,(0 - T_o)}{t}. \tag{13}$$

If a local source (lamps, machinery, a person leaning on it, a leak of warm air) delivers more than q_{melt}, ice starts to melt at a rate

$$v = \frac{q - q_{\text{melt}}}{\rho\, L_f}. \tag{14}$$

Example: an excess of 150 W/m^2 melts ≈ 4.2 cm/d. The time for heat to pass through the full thickness is

$$\tau = \frac{t^2}{\alpha}, \qquad \alpha = \frac{k}{\rho c_p} = 1.87 \cdot 10^{-6}\ \text{m}^2/\text{s}. \tag{15}$$

Table 4: Thermal results, design case ($T_o = -60\,°\text{C}$, $R_{ins} = 4\,\text{m}^2\,\text{K/W}$).

L (m)	t (m)	q (W/m^2)	T_i (°C)	q_{melt} (W/m^2)	τ (days)	Ring 1 heat loss (kW)
16	0.60	19.0	−56.0	288	2.3	85
24	0.60	19.0	−56.0	288	2.3	146
32	1.25	18.0	−52.1	137	9.9	208
40	3.40	15.3	−41.2	49	75.5	246

With $R = 4$ insulation the inner face stays at −56–−41 °C, far from melting. Thicker shells run slightly warmer inside, because the ice itself adds thermal resistance. The values of τ (days to weeks) mean temperature sensors have time to catch a hot spot before any ice melts.

8.3 Insulation and structural safety

Heat from inside does not only risk melting; it warms the ice and speeds up creep. For a 32 m span with $t = 1.25$ m: With too little insulation ($R < 2$), the inner face warms to −35–−7 °C and the buckling safety factor drops below 1: **the arch slowly sags until it collapses**, long before any ice melts. Creep, not melting, is the governing criterion. Thick insulation also cuts the habitat's heating demand.

8.4 Daily and seasonal temperature swings

The penetration depth of a temperature wave is $d = \sqrt{\alpha P/\pi}$:

- daily (1 sol): $d \approx 0.23$ m, only the outer skin;
- seasonal (1 Mars year): $d \approx 6.0$ m, deeper than the shell. The whole shell warms in summer.

Table 5: Sensitivity to insulation (32 m span).

R_{ins} (m² K/W)	q (W/m²)	T_i (°C)	FS buckling	Heat loss (kW)
0.25	106.8	−6.7	0.05	1,234
0.50	81.7	−20.8	0.17	944
1.00	54.8	−34.8	0.58	633
2.00	32.7	−45.4	1.48	378
4.00	18.0	−52.1	2.58	208
8.00	9.5	−55.9	3.45	110

Damaged case: the reflective cover is lost, so the mean outer-face temperature rises to −55 °C with a ±30 °C swing (summer peak −25 °C) for the whole design life. Only the 16 m span stays safe.

Table 6: Damaged case (reflective cover lost).

L (m)	t (m)	T_i (°C)	ε_c (%)	E_{eff} (GPa)	FS	t required (m)
16	0.60	−51.2	0.04	0.121	2.93	0.55
24	0.60	−51.2	0.13	0.054	0.39	1.96
32	1.25	−47.4	0.39	0.024	0.31	> 8
40	3.40	−37.0	1.57	0.007	0.37	> 8

Larger spans require a maintained reflective cover, temperature monitoring, and a repair plan.

9 Thrust at the arch feet

The horizontal force at the feet per metre of width, $H = \rho g t\, a$, is carried by basalt floor ties between the inner and outer feet. For a 32 m span, the force per 2.5 m segment is ≈ 97 kN. With basalt-bar tensile strength $\approx 1{,}000$ MPa and FS 3, the required area is ≈ 300 mm²: about one Ø20 mm bar per segment.

10 Conclusions and design rules

1. Compressive stress is not a problem for any span studied.
2. Thickness is set by creep-driven buckling: 0.6 m (16–24 m), 1.25 m (32 m), 3.4 m (40 m). A 40 m span is impractical without ribs.
3. Ice temperature is the key design variable. Keep the inner face ≤ -50 °C with insulation $R \geq 4$ m² K/W; keep the outer face near −60 °C with a reflective cover.
4. Inner-surface melting only occurs where local flux exceeds q_{melt} (50–290 W/m²). Heat sources must not touch or shine directly on the ice; an air gap and insulation blanket are mandatory.
5. Fit thermistors and strain gauges in every segment. The thermal response time of several days leaves time to act.
6. Spans of 24–32 m are feasible for public spaces (sports hall, pool) if rule 3 is met.

11 Limitations

The buckling model uses an effective-modulus method and two-hinged arch coefficients, not finite-element analysis. Each arch is treated as an independent strip; shell action between segments and ring curvature are ignored, so results are conservative. Creep parameters come from terrestrial glacier ice; ice frozen in bags may differ. Arcadia surface temperatures and the effect of the

reflective cover should be checked with a climate model (Mars Climate Database). Sublimation from the outer face has not been calculated.

References

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